Report Exercise 5
Advanced Computer Architectures

()

Michiel Vandromme & Joffrey Van Cauwenberghe

March 25, 2024



Class: Master Software Engineering

1 Introduction

In this lab, we compared synchronous and asynchronous execution of kernels and memory copies. In synchronous execution, the cpu waits for the gpu's operations before resuming execution. In asynchronous execution, the cpu can continue computing without having to wait for the gpu.

2 Implementation

To test the influence of both executions, we had to create 4 kernels: max, min, sum and product. These operations were performed on the elements of a fixed array of size N. For synchronous execution, we executed each kernel sequentially. We used the cudaDeviceSynchronise method each time after calling the kernel so the CPU waited for the result. (Kernel calls are async by nature)
For asynchronous execution, we used cudaMemcpyAsync, which also allows us to proceed in the CPU while waiting on the GPU.

3 Evaluation & Conclusion

The figure below shows the performance differences of the two processing methods. As we expected, the execution time of the asynchronous version is a lot faster because of the parallel behaviour of the CPU compared to the GPU. This makes sense because in the synchronous version, operations are performed sequentially. In the asynchronous version, a lot of time is saved because the CPU can already start the array generation while the gpu is still processing the previous kernel.

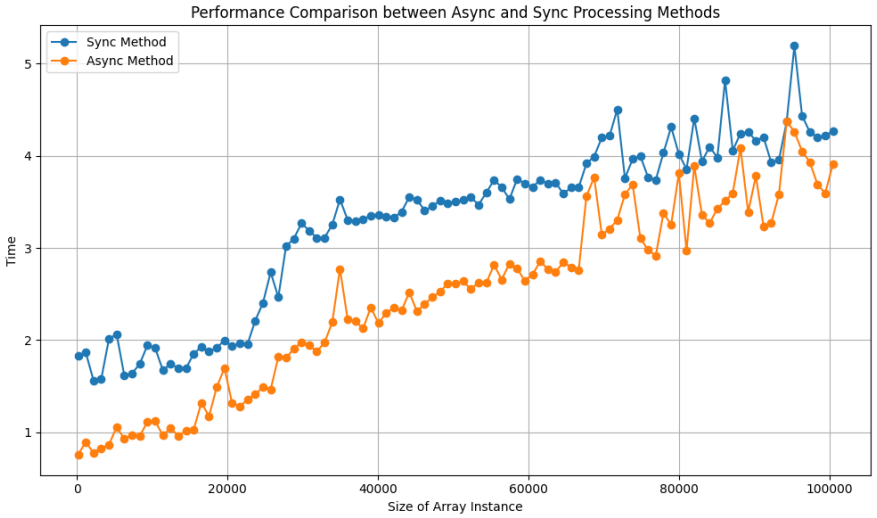


Figure 1: Performance Comparison between Sync & Async Processing